Experiment 5 - Fabry-Pérot Interferometer

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1 Aim

To set up a Fabry-Pérot Interferometer and to determine the wavelength of a mercury lamp and the seperation of the sodium doublet.

2 Background

A Fabry-Pérot consists of two pieces of glass partially coated in a reflective material. The beam of light for analysis is passed through the back of one of the plates and is effectivey trapped between the two plates. At each of the points the light strikes the second mirror, some light is let through.
By placing a lens before and after the glass plates the light can be focused such that all the reflected rays from one source focus onto the same area. This will form an interferance pattern on a screen placed behind it.

2.1 Applications

Fabry-Pérot interferometers are often used to measure the wavelength of a given light beam, by modulating the distance between the two plates.
A version with the mirrors fixed in place, called an etalon, is often used in systems with frequency multiplexing to allow multiple channels to be sent over a single fibre optic cable both directions

Accepted values: Calibration factor may not Applying

3 Method

1. The micrometer was set to 5 mm. The mercury lamp was turned on.
2. When looking down the telescope, an easily identifiable point was chosen and 50 lines passing that point were counted. The micrometer reading was then recorded.
3. This was repeated up until 300 lines passed that point in total.

4 References

1. R. Nave, *Fabry-Perot Interferometer*, The Hyperphysics Textbook, accessed at https://www.hyperphysics.phy-astr.gsu.edu/hbase/hph.html
2. National Institute of Standards and Technology, *https://physics.nist.gov/PhysRefData/Handbook/Tables/mercurytable2.htm*
3. P. Juncar, J. Pinard, J. Hamon, A. Chartier, 1981, *Absolute Determination of the Wavelengths of the Sodium D1 and D2 Lines by Using a CW Tunable Dye Laser Stabilized on Iodine*, Metrologia 17, 77 (1981).